Kelvin Del Castillo

1) 57916 → 0b

57916 | 2
0 28958 | 2
0 14479 | 2
1 7239 | 2
1 3619 | 2
1 1809 | 2
1 904 | 2
0 452 | 2
0 226 | 2
0 113 | 2
1 56 | 2
0 28 | 2
0 14 | 2
0 7 | 2
1 3 | 2
1 1 | 2
1 0

57916 → 0b1110001000111100 *

2) 0xB204 → decimal

0xB204 → 0b 1011 0010 0000 0100 → decimal

$(1\times2^2)+(1\times2^8)+(1\times2^{11})+(1\times2^{12})+(1\times2^{14}) = 4+256+2048+4096+16384 =$ 22788 *

3) 0x 7506 → 0

0x → 0b 0111 0101 0000 0110 → 072406 *

4) 0b 1100110011101,100100 → decimal

0b 1100110011101 ⇒ $(1\times2^0)+(1\times2^2)+(1\times2^3)+(1\times2^6)+(1\times2^7)+(1\times2^9)+(1\times2^{10})+(1\times2^{11}) = 1+4+8+64+128+1024+2048 = 3277$

0b1100110011101 → 3277

Decimal $0.100100 \xrightarrow{-1-2-3-4-5-6} (1\times2^{-1}) + (1\times2^{-4}) = 0.5 + 0.0625$

$0.100100 \rightarrow 0.5625$ *

$0b\ 1100110011101.100100 \rightarrow 3277.5625$ *

5) -8017 Complemento a 2 32 bits → 0x

8017 | 2
1 4008 | 2
0 2004 | 2
0 1002 | 2
0 501 | 2
1 250 | 2
0 125 | 2
1 62 | 2
0 31 | 2
1 15 | 2
1 7 | 2
1 3
3 | 2
1 1 | 2
1 0

$8017 \rightarrow 0b1111101010001$

0b00000000000000001111101010001
Comp 1 11111111111111111000010101110
Com 2 1
11111111111111111000010101111 *

1111 1111 1111 1111 1110 0000 1010 1111 → 0xFFFFE0AF
0xFFFFE0AF *

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6) -4707.357 IEEE-754 32bit → 0x

Signo | Exponente | Mantiz

4707 | 2
1 2353 | 2
1 1176 | 2
0 588 | 2
0 294 | 2
0 147 | 2
1 73 | 2
1 36 | 2
0 18 | 2
0 9 | 2
1 4 | 2
0 2
2 | 1 0
1

4707 → 0b1001001100011

decimal → 0b	
0.357 x2 = 0	0b1001001100011.0101101101
0.714 x2 = 1	Signo: 1
0.428 x2 = 0	Exponente: 12+127 = 139 = 11001011
0.856 x2 = 1	Mantis: 01011011010..
0.712 x2 = 1	
0.424 x2 = 0	1110,0101,1010,1101,1010,0000,0000,0000
0.848 x2 = 1	0xE5ADA000
0.696 x2 = 1	0xE5ADA000 *
0.392 x2 = 0	
0.784 x2 = 1	

11001011

IEEE-754 32bit

7) 0x43DF05A8 → decimal

0b 0100 0011 1101 1111 0000 0101 1010 1000,

Signo: 0

Exponente: $\overset{7\,6\,5\,4\,3\,2\,1\,0}{10000111} \rightarrow (1\times2^{0})+(1\times2^{1})+(1\times2^{2})+(1\times2^{7})=$

$1+4+128=133-127=\underline{6}$ *

Ma...: $\overset{-1\,2\,3\,4\,5\,6\,7\,8\,9\,10\,11\,12\,13\,14\,15\,16\,17\,18\,19\,20\,21\,22\,23}{10111110000010110101000}$

$(1\times2^{-1})+(1\times2^{-3})+(1\times2^{-4})+(1\times2^{-5})+(1\times2^{-6})+(1\times2^{-7})+(1\times2^{-13})+(1\times2^{-15})+(1\times2^{-16})+(1\times2^{-18})+(1\times2^{-23})$

$= 0.5+0.125+0.0625+0.03125+0.015625+0.0078125$

$\underline{0.7421875}$ *

0x43DF05A8 → $\underline{64.7421875}$ *